

4th Multi-modal Aerial View Image Challenge - Translation: Our Solution

Anonymous CVPR submission

Paper ID *****

Abstract

001 This paper presents our solution for the 4th Multi-modal
002 Aerial View Image Challenge - Translation (4th-MAVIC-T),
003 submitted to the PBVS workshop at CVPR 2026. The chal-
004 lenge focuses on multi-modal image translation between
005 SAR, EO, IR, and RGB imagery to enable high-quality and
006 high-fidelity cross-modal image generation. Our approach
007 addresses the domain translation problem by leveraging re-
008 cent advances in remote sensing image-to-image transla-
009 tion, building upon GAN-based methods and diffusion mod-
010 els for SAR-to-optical translation. We reference the 3rd
011 Multi-modal Aerial View Image Challenge [1] as a foun-
012 dation for this work.

013 1. Related Work

014 Image-to-image translation has been extensively studied in
015 computer vision, with foundational works including Cy-
016 cleGAN for unpaired translation [8] and Pix2PixHD for
017 high-resolution conditional generation [6]. Recent ad-
018 vances in remote sensing image-to-image translation have
019 explored various approaches, including GAN-based meth-
020 ods for SAR-to-optical translation [2, 5], conditional dif-
021 fusion models with spatial-frequency refinement [3], effi-
022 cient end-to-end diffusion models for one-step translation
023 [4], and adaptive domain shift techniques in diffusion mod-
024 els [7].

025 References

- 026 [1] Dylan Bowald, Justice Wheelwright, Oliver Nina, Angel
027 Sappa, Riad Hammoud, Erik Blasch, and Nathan Inkawhich.
028 3rd Multi-modal Aerial View Image Challenge: Sensor Do-
029 main Translation - PBVS 2025. In *Proceedings of the*
030 *IEEE/CVF Conference on Computer Vision and Pattern*
031 *Recognition*, pages 4679–4688, 2025. 1
- 032 [2] Xikun Hu, Puzhao Zhang, and Yifang Ban. Gan-based SAR
033 to Optical Image Translation in Fire-Disturbed Regions. In
034 *IGARSS 2022 - 2022 IEEE International Geoscience and Re-*
035 *remote Sensing Symposium*, pages 1456–1459, 2022. 1
- 036 [3] Jiang Qin, Kai Wang, Bin Zou, Lamei Zhang, and Joost
037 van de Weijer. Conditional Diffusion Model With Spatial-

- Frequency Refinement for SAR-to-Optical Image Translation. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–14, 2024. 1 038 039 040
- [4] Jiang Qin, Bin Zou, Haolin Li, and Lamei Zhang. Efficient End-to-End Diffusion Model for One-Step SAR-to-Optical Translation. *IEEE Geoscience and Remote Sensing Letters*, 22:1–5, 2025. 041 042 043 044
- [5] Haixia Wang, Zhigang Zhang, Zhanyi Hu, and Qiulei Dong. SAR-to-Optical Image Translation With Hierarchical Latent Features. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–12, 2022. 1 045 046 047 048
- [6] Ting-Chun Wang, Ming-Yu Liu, Jun-Yan Zhu, Andrew Tao, Jan Kautz, and Bryan Catanzaro. High-Resolution Image Synthesis and Semantic Manipulation With Conditional GANs. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 8798–8807, 2018. 049 050 051 052 053
- [7] Zihao Wang, Yuzhou Chen, and Shaogang Ren. Adaptive Domain Shift in Diffusion Models for Cross-Modality Image Translation, 2026. 1 054 055 056
- [8] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired Image-To-Image Translation Using Cycle-Consistent Adversarial Networks. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2223–2232, 2017. 057 058 059 060 061